

Figure 10: `ifconfig` command on hosts h1 and h2

No.	Time	Source	Destination	Protocol	Length	Info
170	23.776036371	10.0.0.11	10.0.0.12	OpenFlow	182	Type: OFPT_PACKET_IN
173	23.776194755	10.0.0.11	10.0.0.12	OpenFlow	188	Type: OFPT_PACKET_OUT
175	23.777526252	10.0.0.12	10.0.0.11	OpenFlow	182	Type: OFPT_PACKET_IN
178	23.777643123	10.0.0.12	10.0.0.11	OpenFlow	188	Type: OFPT_PACKET_OUT

Figure 11: Stop capturing Wireshark packets

The following are the OpenFlow messages exchanged, as shown in Figure 9.

- **Hello message:** Controller and switch share their version number.
- **Features request:** Controller requests the switch to see the available ports.
- **Features reply:** As a reply, the switch sends a list of ports, their speed, tables and actions to the controller.
- **Set Config:** Controller requests the switch to send flow expirations.
- **Port status:** Switch sends port status to the controller.

Step 6: Initiate a ping between host h1 and host h2, and see the packets shared between them in the Wireshark window by applying the filter 'ICMP' in the filter text box.

Figure 10 shows the ICMP packets shared between h1 and h2.

When we ping from host h1 to host h2, the following types of OpenFlow packets are exchanged.

- **PACKET_IN:** The switch sends this message to the controller when it doesn't find a match in its flow table for a packet received.
- **PACKET_OUT:** The controller sends a packet out to one or more switch ports.

Figure 9 shows the various fields of OpenFlow packets like version number, type of packet sent, length of the packet, transaction ID, port number, etc. It also contains the IP address and MAC address of hosts h1 and h2, which can be verified on the terminal (shown in Figure 10) by running the `ifconfig` command.

Step 7: To stop capturing packets in Wireshark, click the red icon, as shown in Figure 11.

References

- Mininet walkthrough, <http://mininet.org>
- <https://www.wireshark.org/>

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